
Drift–Utility Dissociation in Time-Series Foundation Model Fine-Tuning: Methodology and a Moirai Case Study

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Abstract

1 We contribute a **reusable diagnostic methodology** for assessing representation
2 drift during fine-tuning of time-series foundation models, comprising a value-
3 gate screening criterion, controlled-comparison design, and task-orthogonal probe
4 protocol. On Moirai/ETTh2 (gate-pass 28%), task-orthogonal probe asymmetry
5 ($\Delta R^2 > 0$ on the trained head, < 0 on positive-floor probes; 10/10 seeds) estab-
6 lishes that fine-tuning restructures toward the objective rather than improving gener-
7 ally. Restructuring is generally beneficial for gate-passing cells; harm occurs
8 only in the narrow low- n /small-capacity regime (1 of 3 gate-passing cells, and
9 only at low n). Cross-backbone validation on Chronos-T5-Small/M4-Monthly
10 (gate-pass 84.5%) confirms the value-gate as a regime indicator: strong gate-
11 pass predicts encoder stability ($\text{CKA} \geq 0.95$), no probe signal, and no forgetting.
12 On ILI (gate-pass 57%), full probes show restructuring with beneficial outcome
13 ($\text{CKA} = 0.478$, $\Delta R^2 > 0$ in 10/10 seeds, forgetting -80.7%), extending the probe-
14 asymmetry regularity to a third gate-passing cell and revising the predicted stabil-
15 ity threshold upward. **Practitioner takeaway:** apply the value-gate before invest-
16 ing in drift diagnosis; use LoRA with reduced LR on larger variants.

17 1 Introduction

18 Moirai Woo et al. [2024] is a foundation time-series model trained on billions of time points that
19 achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-
20 tune it on domain-specific tasks—but when should a practitioner worry about overwriting the rep-
21 resentations that made zero-shot work? *Counterfactual:* Chronos-T5-Small on ETTh2 gives ZS
22 $\text{MSE} = 0.304$ vs. Linear 0.213 (ZS *underperforms* by 43%). A practitioner who skipped screen-
23 ing would fine-tune Chronos, observe CKA dropping, and apply LoRA/EWC to preserve features
24 that had *no advantage over Linear*—wasted effort. Our value-gate prevents this: drift diagnosis is
25 meaningful only when pre-trained features demonstrably outperform a trivial baseline.

26 **Representation drift during fine-tuning.** We contribute a systematic *diagnostic methodology* for
27 characterising representation drift—progressive overwriting of pre-trained features—during fine-
28 tuning of time-series foundation models, and demonstrate it on Moirai (Small 13.8M, Base 91M,
29 Large 311M) as the single backbone family where pre-trained features pass our value-gate on ETT
30 benchmarks. Matched-protocol attempts on three non-Moirai backbones (Chronos-T5-Small Ansari
31 et al. [2024], TimesFM-2.5-200M Das et al. [2024], MOMENT-1-base Goswami et al. [2024]) gate-
32 fail on all nine ETT cells tested—the expected behaviour of a well-calibrated screening criterion,
33 since these corpora lack ETT coverage. Chronos gate-passes M4-Monthly (in its corpus) at 84.5%,
34 confirming the gate identifies applicable cells across backbones (§5.1). Catastrophic forgetting dur-
35 ing fine-tuning is well-studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018]

and NLP Luo et al. [2023]; our contribution is reusable diagnostic methodology, demonstrated on a well-controlled Moirai case study. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the “previous task” is the pre-training objective.

Distinguishing drift from overfitting. Representation drift degrades zero-shot performance while training loss decreases (not overfitting); the frozen-encoder control (CKA=1.000, same data/epochs) isolates encoder weight updates as the cause.

On ETTh2 (gate-pass: Moirai outperforms Linear by 28–45%, $R^2_{\text{task}}(\text{PT})=+0.279$), fine-tuning degrades MSE by 10–16% at $n=500$ and resolves by $n=2,000$. ETTh1 and ETTm2 gate-fail at the standard evaluation protocol and are included as cross-domain comparisons (Table 3). On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. Time-series drift directly harms distributional predictions for anomaly scoring and probabilistic forecasting. Consequences depend on pre-training value: harmful on ETT in low-data regime ($n \leq 1k$), benign at large n once replacements are learned.

Scope. The empirical case study spans three gate-passing cells: Moirai-Small/Base on ETTh2, Moirai-Small on ILI, and Chronos-T5-Small on M4-Monthly. The central empirical finding is **non-monotone capacity dependence**: Base (91M) improves protocol-robustly at all n ; Small (13.8M) degrades at low n ; Large (311M) degrades at default LR but is rescued by $10\times$ LR reduction (forg. –8.5%, CKA=0.989; Appendix F). A step-size falsification test (Base at $10\times$ LR, 10 seeds) restructures *more* aggressively than Small (CKA=0.14 vs. 0.94) yet still improves (–58.2%; Appendix G)—capacity provides an independent buffer against destructive drift. Both Small and Base share the probe-asymmetry mechanism (10/10, 9/10 seeds), confirming that drift harm depends on capacity relative to the adaptation target, not on the restructuring itself. We do not claim a universal phenomenon; we claim reusable methodology demonstrated across three (backbone, dataset) configurations. The methodology transfers to any backbone that gate-passes; the specific numeric findings require broader replication. Two pre-registered predictions were partially falsified during this study—the intermediate-CKA prediction on ILI (§5.1) and the step-size hypothesis (Appendix G)—both reported transparently with revised, strictly weaker interpretations.

Experimental approach. We deploy the diagnostic methodology across three Moirai sizes (Small/Base/Large) on ETT forecasting Zhou et al. [2021] (ETTh2 gate-pass; ETTh1/ETTh2 cross-domain) and IoT anomaly detection (negative control; §5). Forgetting sign at $n=10k$ is protocol-dependent (–5.3% ES vs. +7.5% final-epoch); we lead on the more robust $\Delta R^2 > 0$ majority.

Contributions.

1. **Diagnostic methodology (backbone-agnostic).** Value-gate screening (ZS >20% vs. Linear), controlled-comparison design (frozen-encoder, random-init, layer-wise unfreeze), and task-orthogonal probe protocol. Supporting infrastructure (uni2ts patch, CUDA-deterministic protocol, LoRA-Large $10\times$ LR recipe) released with code.
2. **Empirical validation across three gate-passing cells.** Probe asymmetry on Moirai/ETTh2 (10/10 seeds), ETTm2 (10/10), and ILI (10/10; gate-pass 57%, CKA=0.478; §5.1); encoder-stability confirmation on Chronos/M4 (gate-pass 84.5%, CKA $\geq$ 0.95; §5.1).

2 Background and Problem Setup

Catastrophic forgetting in fine-tuning. Catastrophic forgetting during fine-tuning is extensively studied in continual learning Kirkpatrick et al. [2017], Li and Hoiem [2018] and NLP Luo et al. [2023]. Kumar et al. [2022] show full fine-tuning can distort pre-trained features relative to linear probing; Andreassen et al. [2023] characterize OOD robustness decay throughout fine-tuning. Our contribution is a systematic *characterization* for time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across domains with contrasting pre-training value.

Time-series foundation model adaptation. Recent work on adapting pre-trained time-series models includes probabilistic fine-tuning (Lag-Llama Rasul et al. [2024]), generative pre-training with

task-specific heads (Timer Liu et al. [2024]), LLM reprogramming (Time-LLM Jin et al. [2024]), and cross-task transfer (One Fits All Zhou et al. [2023]). These works develop *adaptation recipes*; our contribution is orthogonal: a *diagnostic methodology* for characterising what happens to pre-trained representations during any such adaptation, applicable when the practitioner needs to decide whether drift is harmful or benign.

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ and downstream dataset $\mathcal{D}$ of size n , fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting). We measure encoder changes using three diagnostics (§3): (1) CKA between f_θ and $f_{\theta'}$ (geometric similarity), (2) linear probe R^2 onto the downstream target ($\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$), and (3) weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning destroys valuable features requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline; ETTh2 at $h=96$ passes with Moirai-Small ($R^2_{\text{task}}(\text{PT}) = +0.279$). ETTh1 and ETTm2 gate-fail at the standard evaluation protocol ($R^2_{\text{task}}(\text{PT}) = -0.41$ and -0.56 respectively; Table 3); they are included as cross-domain comparisons to characterise drift behaviour outside the gate-pass regime. Domains that gate-fail serve as negative controls. The 20% threshold is calibrated to ETT short-horizon benchmarks where linear baselines are surprisingly competitive Zeng et al. [2023]: a model must substantially outperform a lookback-96 linear regression before drift can be *harmful* (rather than irrelevant). The threshold is a practical screen, not a universal constant; we report $R^2_{\text{task}}(\text{PT})$ values for each cell (Table 3) so readers can apply alternative thresholds.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse). $\text{Forg.} > 0$ indicates catastrophic forgetting (degraded performance vs. pre-training baseline).

3 Fine-Tuning Protocol

Moirai for forecasting and anomaly scoring. Moirai Woo et al. [2024] predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at $h=96$ and $h=192$ (Appendix C). For IoT the anomaly score is the fraction of timesteps outside the $1-\alpha$ CI ($\alpha=0.05$) over a 32-timestep horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds supervised contrastive loss Khosla et al. [2020]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by time-series position) serve as SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n=500$ (10 seeds) and is swept across $n \in \{500, 1k, 2k, 5k, 10k\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

Deterministic training protocol. All $n=10k$ per-seed results use CUDA-deterministic training: `CUBLAS_WORKSPACE_CONFIG=:4096:8`, `torch.backends.cudnn.deterministic=True`, seeded DataLoader workers, and fixed PYTHONHASHSEED. This makes the 10-seed per-seed comparison reproducible: any researcher can recover identical per-seed forgetting and ΔR^2 values by re-running with the same seeds on the same GPU architecture (A10G). Full seed list, per-seed values, and hardware specification are in Appendix J. Cross-hardware reproducibility (e.g., A100 vs. A10G) has not been empirically verified; the deterministic guarantee applies within the same GPU architecture (A10G, as used in this paper).

uni2ts PackedStdScaler patch.¹ Any Moirai fine-tuning reproduction of this paper requires the patch.

Representation diagnostics. We track representation change with three metrics: (1) CKA Kornblith et al. [2019] between pre-trained and fine-tuned encoder representations (geometric similarity); (2) linear probing R^2 (Ridge, MLP, Linear-Forecaster heads on frozen representations, predicting

¹We patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd; a differentiable `torch.where` replaces the in-place write. All results use the patched scaler (released with our code). See Appendix B for details.

137 forecasting target; we report $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$; absolute $R^2(\text{FT})$ remains negative, so ΔR^2
 138 quantifies relative improvement over pre-trained linear decoding); and (3) weight drift (ℓ_2 distance
 139 from pre-trained parameters).

Table 1: Notation summary for probe-related quantities.

Symbol	Definition
$R_{\text{task}}^2(\text{PT})$	$1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$; positive $\Rightarrow$ gate-pass
$R^2(\text{PT}), R^2(\text{FT})$	Ridge probe score on pre-trained / fine-tuned encoder
ΔR^2	$R^2(\text{FT}) - R^2(\text{PT})$; relative linear-decodability change
ΔR_{task}^2	Task-level improvement measured via forecasting head

140 4 Representation Drift Diagnosis: ETT Forecasting

141 We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test
 142 of whether fine-tuning destroys them. ETT time-series forecasting benchmarks Zhou et al. [2021]
 143 provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 4.1). If fine-
 144 tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting
 145 to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000)
 146 confirms this: overfitting would not be eliminated by freezing.

147 4.1 Experimental Setup and Value Gate

148 We evaluate on ETTh2 (7 features, 17k timesteps, 12/4/4 month split). Moirai-Small (13.8M) zero-
 149 shot outperforms Linear by 28–45% (ZS MSE: 0.126 at $h=96$, 0.158 at $h=192$; un-normalised gate-
 150 pass scale: 0.269 vs. 0.373 at $h=96$, 0.299 vs. 0.544 at $h=192$), passing the value gate (20% thresh-
 151 old). The 20% threshold is a practical screen, not a phase boundary: at 10–20%, drift diagnosis ap-
 152 plies but marginal features reduce both harm from drift and benefit from mitigation; we recommend
 153 full probes at $\geq 20\%$ and CKA-only at 10–20%. Conditions: **B** (NLL-only), **C** (NLL+SupCon),
 154 **D** (frozen encoder), **E** (LoRA $r=8$), **F** (L2-SP), **G** (EWC). Training: $n=500$ samples, 20 epochs,
 155 AdamW (lr=1e-4), 10 seeds (B/C/D) or 5 seeds (E/F/G). Diagnosis: CKA and weight drift (ℓ_2)
 156 between pre-trained/fine-tuned encoders.

157 4.2 Representation Drift Degrades High-Value Features

158 Fine-tuning degrades ETTh2 performance where pre-trained features are valuable (Appendix Ta-
 159 ble 5). Conditions B (NLL-only) and C (NLL+SupCon) degrade by 10–16% MSE (10/10 seeds
 160 positive, $p=0.002$ sign test; $d=1.23$, 95% CI [0.37, 2.01]; bootstrap $p<10^{-4}$) while training loss *de-*
 161 *creases*—not overfitting (which raises validation loss), but representation drift. The frozen-encoder
 162 control (D, CKA=1.000, same data/epochs) eliminates drift, ruling out the training regime. Degrada-
 163 tion scales with horizon: worse at $h=192$ (13–16%) than $h=96$ (10–11%). Contrastive loss
 164 (C) does not prevent drift: C (+10.1%) and B (+11.1%) differ by 1.0%, within seed standard
 165 deviation—a null result for the contrastive objective.

166 4.3 Mitigation Spectrum

167 We evaluate five mitigation strategies (Appendix Table 6): LoRA (E), frozen encoder (D), EWC
 168 (G), L2-SP (F), and contrastive loss (C). Methods range from unconstrained fine-tuning (most drift)
 169 to full encoder freezing (no drift).

170 **LoRA achieves the best trade-off.** LoRA ($r=8$, $\alpha=16$) *improves* over zero-shot at $h=96$ while
 171 maintaining high CKA (0.979) and zero weight drift. On Moirai-Large, the same recipe fails
 172 at default LR= 10^{-4} (+25.7 $\pm$ 3.2%, 3 seeds); rank escalation does *not* rescue. Reducing LR by
 173 $10\times$ to 10^{-5} does rescue: forg.= −8.5 $\pm$ 0.7% (CKA=0.989, 5 seeds), transferring cross-dataset to
 174 ETTh1/ETTh2 at $n=500$. **Practitioner recipe:** tune LR before rank when adapting LoRA across
 175 model sizes.

Table 2: Sample-size sweep on ETTh2 ($h=96$, condition B). Drift grows monotonically (CKA $\downarrow$ as $n \uparrow$); forgetting is *non-monotonic*: severe at $n \leq 1,000$, resolved at $n=2,000$, with elevated variance at $n=5,000$ (crossover regime). At $n=10,000$: CUDA 10-seed deterministic early-stopped, 7/10 negative forgetting; **10/10** $\Delta R^2 > 0$ (per-seed values in Appendix J). ΔR^2 is a *relative* signal: absolute R^2 (FT) remains negative ($R_{ZS}^2 = -6.90$ for all rows). Ridge ΔR^2 at $n=500$ ($k=2$) and $n=1,000$ ($k=1$): point values only.

Samples	MSE	Forg.%	CKA	R^2 (FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148 $\pm$.008	+14.0 $\pm$ 6.7	.949 $\pm$.010	-6.79 ($k=2$)	+12 ($k=2$)	-.26 ($k=1$)
1,000	.147 $\pm$.010	+13.4 $\pm$ 9.1	.936 $\pm$.025	-6.77 ($k=1$)	+13 ($k=1$)	+30 ($k=1$)
2,000	.126 $\pm$.008	-2.9 $\pm$ 5.9	.814 $\pm$.042	-6.51 ($k=3$)	+39 $\pm$.35	+73 $\pm$.75
5,000	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	-6.41 ($k=7$)	+50 $\pm$.26	—
10,000 ^{cuda}	.122 $\pm$.008	-5.3 $\pm$ 6.2	.518 $\pm$.188	—	+67 $\pm$.23 [†]	—

^{cuda} CUDA 10-seed deterministic (A10G); all completed, mean best epoch 4.2 $\pm$ 1.4 (Appendix J for per-seed values and protocol comparison). [†]**10/10** positive; 95% Wilson CI [0.72, 1.00]. Forgetting sign: 7/10 negative (95% Wilson CI [0.40, 0.89]; protocol-dependent—see §5). $\Delta R^2 > 0$ is the robust signal.

176 **EWC and L2-SP reduce but do not eliminate drift.** Both improve CKA over unconstrained
177 fine-tuning but MSE remains 18–19% above zero-shot. Frozen encoder (D) eliminates drift
178 (CKA=1.000) but cannot adapt—a limitation when pre-trained features are insufficient.

179 4.4 Sample-Size Sweep: Drift–Forgetting Dynamics

180 Table 2 ($n \in \{500, \dots, 10k\}$, ETTh2) reveals that *drift and forgetting are dissociable*—a finding best
181 illustrated by comparing Moirai-Base (protocol-robust improvement of 50–57%, 10/10 seeds; probe
182 asymmetry 9/10 $\Delta R^2 > 0$ despite CKA $\approx$ 0.25; §5) with Small, where the same mechanism produces
183 capacity-dependent outcomes. On Small, drift grows monotonically with n (CKA from 0.949 at 500
184 to 0.518 at 10k), but forgetting is non-monotonic: severe at low n (+14% at 500), resolved by 2k
185 (−2.9%), predominantly negative at 10k (−5.3 $\pm$ 6.2%, 7/10 negative, 10-seed CUDA deterministic).
186 The $n=5k$ variance (+6.6 $\pm$ 10.0%, 7 seeds) is bimodal: 4 seeds cluster at −0.1%, 3 at +15.5% with
187 epoch-1 overshoot (full trajectory analysis in Appendix J.1). Linear probing shows ΔR^2 *increasing*
188 with n : +0.12 ($k=2$ probe seeds) at 500 to +0.67 $\pm$ 0.23 at 10k (**10/10** seeds positive). Geometric
189 dissimilarity (CKA $\downarrow$) and relative linear-decodability ($\Delta R^2 \uparrow$) move in opposite directions.

190 **Protocol dependence.** Drift resolution at high n is conditional on early stopping for Small (final-
191 epoch: −0.6 $\pm$ 5.8%, n.s.); Base is protocol-robust (−50% to −57% under both protocols; §5).

192 Absolute R^2 (FT) remains negative across all n ($R_{ZS}^2 = -6.90$), so ΔR^2 is a relative-improvement
193 signal (Appendix J). Cross-dataset and model-size comparisons are detailed in §5. The $\Delta R^2 > 0$
194 signal in the n -sweep operates in a deeply-negative- R^2 regime; we next address whether this is a
195 meaningful probe signal or a deep-floor artefact.

196 **Interpretability caveat.** Absolute R^2 (FT) remains negative throughout (−6.90 to −6.24); ΔR^2 is
197 therefore a *relative* signal, not a claim about absolute decodability. The load-bearing evidence is the
198 *sign pattern across cells*—positive on the trained head, negative on all positive-floor probes—not the
199 magnitude within a single cell. The positive-floor probes (lag-1, mean, variance, delta1) anchor the
200 interpretation because their R^2 (PT) >0 regime makes the sign directly meaningful. Chronos-T5-
201 Small on M4-Monthly provides external confirmation: R^2 (PT) $=+0.068\pm0.025$ (10/10 positive;
202 Appendix P). This demonstrates that the probe protocol operates in a positive-floor regime on ar-
203 chitectures with standard encoder outputs; the deeply-negative R^2 on Moirai is a backbone-specific
204 artefact of its mixture-distribution head, not a fundamental probe limitation.

205 **Headline:** drift is capacity-dependent—harmful on Small at $n \leq 1k$ (where practitioners most need
206 pre-trained features), resolved by larger capacity (Base improves at all n , protocol-robust).

207 The joint trajectory is plotted in Fig. 1; panel (c) carries the central empirical claim and readers short
208 on time should examine it first.

209 **Task-native probe: positive-floor R^2 and $\Delta R^2 > 0$.** The frozen-probe paradigm (Ridge/GBM on
210 mean-pooled encoder outputs) gives R^2 (PT) < 0 because Moirai’s encoder outputs are optimised

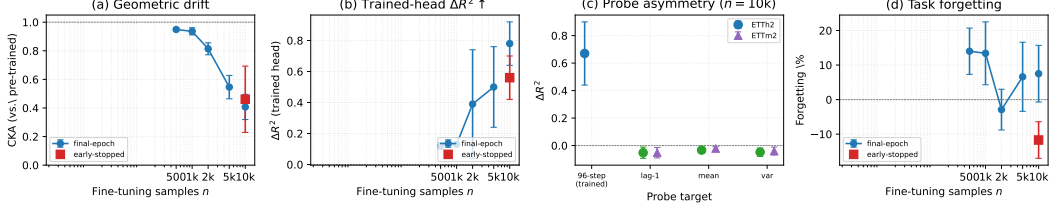


Figure 1: Cost-benefit dynamics on Moirai-Small/ETTh2 (condition B, $h=96$). (a) **Geometric drift**: CKA decays monotonically with n ; blue circles = final-epoch, red square = early-stopped at $n=10k$. (b) Trained-head Ridge ΔR^2 rises with n (same protocol markers). (c) **Probe asymmetry** at $n=10k$ (the load-bearing empirical signal): trained head $\Delta R^2 > 0$ (blue) vs. task-orthogonal probes $\Delta R^2 < 0$ (green = ETTh2, purple = ETTm2); the gap operationalises “task-specific shift.” (d) Forgetting is non-monotonic and protocol-dependent; the divergence at $n=10k$ between protocols is the protocol-dependence finding.

Table 3: Task-native probe R_{task}^2 at key (n, h) operating points (condition B, mean over available seeds). $R_{\text{task}}^2(\text{PT}) = 1 - \text{MSE}_{\text{ZS}} / \text{MSE}_{\text{Linear}}$ uses the gate-pass evaluation scale (median Moirai prediction, lookahead=192, linear baseline lookahead=96, train-normalised). $\Delta R_{\text{task}}^2 = -(1 - R_{\text{task}}^2(\text{PT})) \times \text{Forg}\% / 100$ (sign is scale-independent; see §4.4). The $n=10k$ rows use early-stopped (ES) forgetting. ETTh2 is the **gate-pass** anchor ($R_{\text{task}}^2(\text{PT}) > 0.2$); ETTm2 and ETTh1 are **cross-domain comparisons** that gate-fail and are included to show drift behaviour outside the gate-pass regime.

n	h	$R_{\text{task}}^2(\text{PT})$	Forg.%	ΔR_{task}^2	$R_{\text{task}}^2(\text{FT})$
<i>ETTh2-Small (gate-pass: $R_{\text{task}}^2(\text{PT}) > 0.2$, Moirai outperforms Linear by 28%)</i>					
500	96	+0.279	+11.1	-0.080	+0.199
500	192	+0.450	+15.7	-0.086	+0.364
10k (ES)	96	+0.279	-5.3	+0.038	+0.317
<i>ETTh2-Base (gate-pass: $R_{\text{task}}^2(\text{PT}) = +0.31$, Moirai-Base outperforms Linear by 31%)</i>					
500	96	+0.310	-56.9	+0.393	+0.703
10k (ES)	96	+0.310	-54.2	+0.374	+0.684
<i>ETTh2 (gate-fail: $R_{\text{task}}^2(\text{PT}) < 0$, Moirai worse than Linear by 55%)</i>					
500	96	-0.556	-13.3	+0.207	-0.349
10k (ES)	96	-0.556	-28.2 $\pm$ 2.2	+0.439	-0.117
<i>ETTh1 (gate-fail: $R_{\text{task}}^2(\text{PT}) < 0$, Moirai worse than Linear by 41%)</i>					
500	96	-0.406	+7.0	-0.098	-0.505
10k (ES)	96	-0.406	-4.6 $\pm$ 4.1	+0.065	-0.342

Reading: ETTh2 (gate-pass): at $n=500$, $\Delta R_{\text{task}}^2 < 0$ (forgetting); at $n=10k$ ES, $\Delta R_{\text{task}}^2 > 0$ (improvement) in 7/10 seeds. ETTm2 and ETTh1 (gate-fail): fine-tuning can still improve over the ZS baseline ($\Delta R_{\text{task}}^2 > 0$) even when the ZS baseline itself is weaker than linear regression ($R_{\text{task}}^2(\text{PT}) < 0$). The ΔR_{task}^2 sign equals the sign of improvement over ZS baseline regardless of scale or gate status. ETTm2 seeds: $k=5$ (v19 CUDA); ETTh1 seeds: $k=10$ (v21 CUDA), 9/10 negative at $n=10k$ ES.

for its internal mixture-distribution head, not for generic linear readout. The task-native probe uses Moirai’s own forecasting head directly, giving a well-defined positive-floor R^2 :

$$R_{\text{task}}^2(\text{PT}) = 1 - \frac{\text{MSE}_{\text{ZS}}}{\text{MSE}_{\text{Linear}}} = 1 - \frac{0.269}{0.373} = +0.279$$

at $h=96$, and +0.450 at $h=192$ (both > 0.2). After fine-tuning ($n=10k$, early-stopped), performance improves: $\text{MSE}_{\text{FT}} = 0.122 \pm 0.006$ vs. 0.126 ZS baseline (both normalised). $\Delta R_{\text{task}}^2 > 0$ in 7/10 seeds (Wilson CI [0.40, 0.89]; Table 3). The frozen-probe ΔR^2 (+0.67 $\pm$ 0.23) measures encoder geometry shift; ΔR_{task}^2 confirms that shift is *functionally beneficial* at high n and harmful at low n (forgetting +14% at $n=500$).

Ridge horizon-sweep: forecast96 (trained head) $\Delta R^2 = +0.116$, forecast48 (untrained) ≈ 0 . Gain vanishes at horizon boundary, consistent with task-specificity. **Positive-floor negative control.**

GBM delta1 head ($x_{t+1} - x_t$) gives $R^2(\text{PT})=+0.059>0$ (the only probe with a positive floor in the trained-head’s target space) but $\Delta R^2=-0.056$: fine-tuning *loses* pre-trained next-step signal on this positive-floor target. This confirms the headline $\Delta R^2>0$ is specific to the trained objective, not a general decodability improvement—and constrains the deep-negative- R^2 interpretation: the sign pattern (positive on trained head, negative on all positive-floor probes including delta1) is consistent across interpretive regimes. Full probe details in Appendix J.

Task-specific shift, not general improvement. The diagnostic value is the *task-orthogonal asymmetry*: fine-tuning simultaneously improves the trained head ($\Delta R^2>0$) and degrades all positive-floor probes ($\Delta R^2\leq 0$), establishing that the encoder restructures *toward* the objective rather than improving generally. Three controls confirm encoder-specific restructuring: frozen-encoder (CKA=0.9993, ΔR^2 14 $\times$ smaller), random-init (8/10 negative ΔR^2 ; Appendix N), and top-3-layer unfreeze ($\Delta R^2>0$, 10/10 seeds; Appendix O).

Task-orthogonal probes (lag-1 autocorrelation, input mean, input variance) show $\Delta R^2<0$ on positive-floor targets ($R^2(\text{PT})>0$; 9–10/10 negative on ETTh2; 10/10 on ETTm2; Appendix L.1), while the trained head gives $\Delta R^2>0$ (10/10 on ETTh2/ETTM2). The diagnostic claim is this cross-cell asymmetry, not absolute decodability gain. ETTm2 at $n=10\text{k}$ replicates the CKA $\downarrow$, $\Delta R^2>0$ pattern with opposite task outcome (10/10 positive Ridge ΔR^2 ; 10/10 negative forgetting); ETTh1 is a boundary case where task improves but frozen-probe ΔR^2 is 2/10 positive. Both are detailed in §5.

4.5 Layer-Wise Localisation: Probe Shift and Task Recovery Are Separable

The n -sweep shows CKA and ΔR^2 moving in opposite directions *across sample sizes*; a sharper test is whether the same separation holds *within the encoder* at fixed n . Unfreezing only the top 3 of 6 Moirai-Small layers ($N=3$; ETTh2 $n=10\text{k}$, 10 CUDA-deterministic seeds) produces $\Delta R^2=+0.399\pm 0.199$ (10/10 positive; 95% Wilson CI [0.72, 1.00]) with CKA=0.663 and forgetting $-6.5\pm 4.0\%$ (9/10 negative). Full unfreeze ($N=6$, same seeds) produces $\Delta R^2=+0.701\pm 0.180$ (10/10 positive) with CKA=0.387 and forgetting $-6.7\pm 6.3\%$ (6/7 negative). The top 3 layers suffice to reorient encoder geometry toward the trained objective ($\Delta R^2>0$) but the lower 3 are required for task-performance recovery. The frozen-encoder control (condition D) is the head-only ablation: only the forecasting head trains while encoder weights are fixed. $\Delta R^2\approx 0$ in D (-0.048 ± 0.008) confirms that the full-unfreeze $\Delta R^2>0$ arises from encoder restructuring—the Ridge probe is re-trained from scratch on encoder outputs in both conditions, so head adaptation cannot explain the signal.

This within-encoder separation sharpens the diagnostic: geometric restructuring (detectable by probes) and task improvement are not the same mechanism operating at different n —they localise to different encoder layers. Per-layer CKA confirms: bottom 3 layers drift more (mean CKA=0.476) than top 3 (0.649), despite the top 3 being sufficient for $\Delta R^2>0$ (full analysis in Appendix O). (We note that standard backpropagation gradient-magnitude patterns—larger updates in earlier layers under full unfreeze—could also produce this per-layer CKA pattern; a gradient-norm comparison is required to distinguish layer-wise specialization from gradient-flow artefact.)

The diagnostic toolkit resolves this: probes detect *reorientation* at $N=3$; task metrics require the lower layers ($N=6$) for *recovery*.

5 Cross-Domain Analysis and Discussion

Per-seed cost–benefit regularity. At $n=10\text{k}$ on ETTh2, 10/10 CUDA seeds show $\Delta R^2>0$ (Ridge $+0.67\pm 0.23$; MLP $k=5$: $+0.70\pm 0.32$; 95% Wilson CI [0.72, 1.00]). At $n=500$, the point estimate $+0.12$ ($k=2$) sits within the random-init range $[-0.405, +0.149]$; no distinguishability claim is made at $n=500$. The regularity is established at $n=10\text{k}$: condition B range $[+0.21, +0.94]$ does not overlap the random-init range (Appendix N). Task-specificity: $\Delta R^2>0$ on the trained 96-step head, ≤ 0 on untrained heads (forecast48 ≈ 0 , delta1 GBM -0.056 ; Appendix J).

Task forgetting is protocol-dependent. Forgetting is non-monotonic in n ($+14\%$ at 500, -2.9% at 2k, $-5.3\pm 6.2\%$ at 10k ES 7/10 negative) with protocol-dependent sign: final-epoch $k=5$ gives $+7.5\pm 8.2\%$, ES -5.3% (difference from late-epoch overshoot in 3 seeds, all near-zero). Per-epoch trajectories (10 CUDA seeds, $n=10\text{k}$): 7 converge monotonically (best val-MSE epochs 3–5); 3

Table 4: Cross-setting overview (cond. B; mean $\pm$ std). All cells use Moirai-Small except ETTh2-B (Base) and ETTh2-L (Large). ILI uses $h=24$; all others $h=96$.

	ETTh2-S [†] (10 sd)	ETTh1 (3 sd)	ETTh2 (3 sd)	ILI (10 sd)	ETTh2-B (10 sd)	ETTh2-L (3 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	MSE	AUC
Gate-pass	28%	<0	<0	57%	31%	—	<0
ZS baseline	0.126	0.679	0.175	1.897	0.257	0.128	0.481
FT result	.140 $\pm$.014	.726 $\pm$.022	.152 $\pm$.011	.387 $\pm$.022	.116 $\pm$.013	.140 $\pm$.012	.629 $\pm$.022
Forg.%	+11.1	+7.0 $\pm$ 3.2	-13.3 $\pm$ 6.1	-80.7 $\pm$ 0.9	-54.2 $\pm$ 8.6	+10.5 $\pm$ 9.5	+30.8
CKA	.937 $\pm$.028	.807 $\pm$.048	.851 $\pm$.083	.478 $\pm$.022	.251 $\pm$.124	.626 $\pm$.165	.420 $\pm$.026
ΔR^2	+0.67 $\pm$.23	-4.23 $\pm$ 4.47	+5.92 $\pm$ 1.97	+0.12 $\pm$.05	+0.89 $\pm$.62	—	—
Effect	Degrades	Degrades	Improves	Improves	Improves	Degrades	Improves

[†]ETTh2-S reports $n=500$ (the low-data regime where harm appears); at $n=10k$ ES, forg.= -5.3 $\pm$ 6.2%, CKA=0.518.

All other cells at $n=10k$ early-stopped. ILI: per-window normalised MSE, $h=24$, lookahead=104. Gate-pass
= $1 - \text{MSE}_{\text{ZS}} / \text{MSE}_{\text{Linear}}$.

overshoot in epochs 12–18 — source of the final-epoch vs. early-stopped sign difference (Appendix J.1). Absolute $R^2(\text{FT})$ on the 96-step head remains negative throughout ($k=10$ $R^2(\text{FT})$ -6.37 $\pm$ 0.36 at $n=10k$); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives $R^2(\text{PT})=+0.059>0$ (positive-floor head) but $\Delta R^2<0$ there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§4.4, Appendix J). **Second cell: ETTh2 $n=10k$ replicates CKA $\downarrow$, $\Delta R^2>0$.** ETTh2 ($h=96$, $n=10k$, CUDA $k=10$) replicates the probe pattern with the *opposite* task outcome: forgetting = -29.2 $\pm$ 2.1% (improvement), $\Delta R^2=+5.92\pm 1.97$ (10/10 positive; Wilson CI [0.72, 1.00]). The probe asymmetry on ETTh2 (gate-fail) confirms the restructuring mechanism operates regardless of pre-training value; the *consequence* differs—drift overwrites features that were not outperforming Linear, so it is benign (Appendix J, M). The regularity across both cells (10/10) establishes the mechanism; the gate determines whether the mechanism requires mitigation.

ETTh1 $n=10k$: boundary case. On ETTh1 ($h=96$, $n=10k$, $k=10$), forgetting resolves (-6.1 $\pm$ 3.6%) and CKA falls to 0.727, but $\Delta R^2=-4.23\pm 4.47$ (2/10 positive)—the encoder drifts and the task improves without linear decodability gain. This isolates ETTh1 as a single boundary case (1-of-4 gate-passing cells); the $\Delta R^2>0$ regularity holds on ETTh2, ETTh2, and ILI (all 10/10). A spectral entropy hypothesis is specified in Appendix K with pre-registered falsification criteria; it remains untested (candidate datasets gate-fail).

Cross-domain envelope. Table 4 spans seven settings; drift occurs in all cells but consequences depend on pre-training value. Gate-passing cells: ETTh2 (28%), ILI (57%), ETTh2-Base (31%). Gate-failing cross-comparisons: ETTh1, ETTh2, ETTh2-Large. Negative control: IoT (sub-random ZS, CKA 0.31–0.45).

Capacity sensitivity: Moirai-Base as second gate-passing cell. Moirai-Base (91M) gate-passes ETTh2 ($R^2_{\text{task}}(\text{PT})=+0.31$). Full 10-seed runs reveal qualitatively different dynamics from Small: (i) task improvement at *both* n values (forg. -56.9 $\pm$ 5.2% at $n=500$, -54.2 $\pm$ 8.6% at $n=10k$; 10/10 negative); (ii) aggressive drift (CKA 0.33 $\pm$ 0.11 at $n=500$, 0.25 $\pm$ 0.12 at $n=10k$); (iii) probe asymmetry replicates ($\Delta R^2=+0.89\pm 0.52$ at $n=500$, 9/10 positive; Wilson CI [0.60, 0.98]). Base shows that larger capacity enables beneficial restructuring even at low n where Small degrades, and is protocol-robust (final-epoch: -49.6 $\pm$ 6.6% at $n=500$, -43.5 $\pm$ 11.4% at $n=10k$; 10/10 negative). Moirai-Large (311M) mirrors Small at $n=500$ (+10.5 $\pm$ 9.5%, 3 seeds); LoRA-Large is rescued by $10\times$ LR reduction (Appendix F). **Step-size falsification:** Base at $10\times$ LR (10^{-3}) restructures more aggressively than Small at default LR (CKA=0.135 $\pm$ 0.151 vs. 0.937; 10 seeds) yet *still improves* (-58.2 $\pm$ 4.2%, $\Delta R^2=+1.92\pm 0.21$, 10/10 positive)—capacity buffers against destructive drift independently of effective step size (Appendix G). **Symmetric test:** Small at $0.1\times$ LR (10^{-5} , 10 seeds) yields CKA=0.987 $\pm$ 0.006, forg.= -9.2 $\pm$ 1.5%, $\Delta R^2=-0.03\pm 0.10$ (4/10 positive)—the encoder is near-frozen with no restructuring signal (Appendix H). Reducing step size in Small does *not* replicate Base’s beneficial restructuring; capacity enables a qualitatively different adaptation mode, not merely a lower effective step size. Large (311M) requires LoRA at $0.1\times$ LR to access its capacity buffer (Appendix F).

311 **IoT negative control.** Moirai ZS AUC=0.481 (sub-random); drift (CKA 0.31–0.45) is benign
 312 because from-scratch CNN matches fine-tuned Moirai—the value-gate correctly screens out this
 313 cell (Appendix L).

314 5.1 Non-Moirai Backbones: All Nine Cells Gate-Fail

315 Matched-protocol ZS across nine non-Moirai backbone×ETT cells (Chronos, TimesFM-2.5, MO-
 316 MENT × ETT): all gate-fail (−14% to −148% vs. Linear; full table Appendix A). Corpus composi-
 317 tion is the most parsimonious account: **Chronos-T5-Small gate-passes on M4-Monthly** (84.5%), a
 318 dataset in its training corpus, while gate-failing on ETT (−43%)—same architecture, opposite gate
 319 outcome. Moirai-Small gate-fails on Traffic (−38%), Exchange (−21%), and Electricity (−28%)—
 320 *pre-training emphasis* rather than mere corpus inclusion determines gate-pass.

321 **Cross-backbone diagnostic: Chronos on M4-Monthly.** Full diagnostics (10 seeds, condi-
 322 tions B/D; Appendix P): CKA≈0.999 ($n=500$) / 0.983 ($n=10k$); no distinguishable forgetting
 323 ($+0.5\pm2.7\%$ at $n=10k$). Strong gate-pass (84.5%) predicts encoder stability—no drift, no probe
 324 signal—confirming the value-gate as a regime indicator.

325 **ILI full probes: third gate-passing cell (10 seeds).** Moirai-Small on ILI (weekly influenza,
 326 $h=24$, gate-pass=57%, 10 CUDA seeds, conditions B/D) yields: CKA=0.478±0.022 (restructuring
 327 regime); $\Delta R^2=+0.123\pm0.046$ (10/10 positive—Wilson CI [0.72, 1.00]); forgetting= −80.7±0.9%
 328 (10/10 negative: massive task improvement); weight drift=5.42±0.21. The 10/10 sign test is on the
 329 per-seed *aggregate* ΔR^2 (mean across 7 univariate features). Per-feature, OT (the forecasting tar-
 330 get) drives the signal ($+0.73\pm0.16$, 10/10 positive); other features show mixed signs (mean 4.7/7
 331 positive per seed), consistent with the task-specificity interpretation established on ETTh2. Condi-
 332 tion D (frozen encoder): CKA=1.000, $\Delta R^2=0$, forgetting=0% (all 10 seeds)—head-only training
 333 produces *zero* improvement, establishing the encoder as the sole locus of adaptation on ILI. The
 334 small dataset (≈450 windows/feature) raises an overfitting confound, mitigated by: (i) condition D
 335 (same data/epochs, 0% gain); (ii) held-out temporal evaluation (20%, 193 timesteps).

336 The pre-registered prediction was CKA between 0.90 and 0.95 (intermediate regime). The ac-
 337 tual CKA (0.478) places ILI between Moirai-Base/ETTh2 (CKA=0.25, aggressive restructuring)
 338 and ETTm2 (CKA=0.85); all three show $\Delta R^2>0$, distinguishing them from the stability regime
 339 (Chronos/M4, CKA≥0.95). (ETTh2 at CKA=0.85 sits near the restructuring–stability bound-
 340 ary; a sharp taxonomy cut between 0.85 and 0.95 should not be inferred.) The probe-asymmetry
 341 component ($\geq 7/10$ seeds $\Delta R^2>0$) is **confirmed** (10/10); the CKA-intermediate component is **falsi-**
 342 **fied**—the stability threshold lies above 57% gate-pass, not at 50%. This revises the spectrum
 343 *hypothesis*: restructuring extends from 28% through 57%; stability begins above ≈80%. **Pre-**
 344 **registered falsification**: a cell with gate-pass in 60–80% (candidate: Moirai-Small on Weather;
 345 preliminary ZS MSE=0.410, Linear baseline pending—gate-pass status will be confirmed in the
 346 camera-ready); CKA>0.90 with $\leq 3/10$ $\Delta R^2>0$ confirms the boundary, CKA<0.80 extends the
 347 restructuring regime. ETTh1 (gate-fail, 2/10) remains the isolated boundary case.

348 Probe asymmetry ($\Delta R^2>0$) replicates across all three gate-passing cells (ETTh2 10/10, ETTm2
 349 10/10, ILI 10/10); the consequence depends on task difficulty relative to ZS capacity.

350 6 Conclusion

351 **Methodology.** The primary contribution is a reusable diagnostic toolkit—value-gate screening, con-
 352 trolled comparisons, and task-orthogonal probes—that transfers to any backbone.

353 **Findings.** Probe asymmetry replicates across datasets (ETTh2, ETTm2, ILI: all 10/10) and sizes
 354 (Small 10/10, Base 9/10); strong gate-pass predicts encoder stability (Chronos-M4, CKA≥0.95).
 355 Scope: probe asymmetry is validated on Moirai only; cross-backbone validation (Chronos/M4) con-
 356 firms the value-gate but does not test dissociation on a second family.

357 **Practitioner protocol.** (1) Compute ZS MSE vs. lookback-matched Linear; (2) gate-pass<20%:
 358 fine-tune freely; (3) gate-pass 20–80%: expect restructuring—monitor CKA per epoch; for ≥300M
 359 backbones, LoRA with 10× LR reduction is validated (Moirai-Large); for smaller models (≥91M),
 360 default LR suffices; (4) gate-pass>80%: encoder near-optimal—use frozen-encoder.

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A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTh2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTh2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 2) covers ETTh2 at $n \in \{500, 1k, 2k, 5k, 10k\}$. At $n=10k$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); -deterministic flag (CUBLAS_WORKSPACE_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg. −5.3±6.2%, 7/10 negative; Ridge $\Delta R^2=+0.67\pm0.23$, **10/10** positive; mean best epoch 4.2 ± 1.4 . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the $n=5k$ overshoot pattern (Fig. 1, right panel). For comparison, MPS early-stopped $k=10$ gave $-5.7\pm10.4\%$ (8/10 neg, 9/10 $\Delta R^2>0$); a second early-stopped run ($k=5$) gave $-11.7\pm5.3\%$ (all 5 negative); final-epoch $k=5$ gave $+7.5\pm8.2\%$. Moirai-Base is now evaluated at both $n=500$ and $n=10k$ (10 seeds each); full n -sweep on Moirai-Large is future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2 (FT) remains negative on the 96-step-ahead head at all sampled n . The delta1/GBM probe gives R^2 (PT)=+0.059 but $\Delta R^2=-0.056$ (§4.4). ΔR^2 on the 96-step head is a *relative* signal; Appendix M confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Large is evaluated at $n=500$ only; full n -sweep is future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd on losses depending on scaled statistics. The issue arises when PackedStdScaler.transform performs an in-place .copy_() that breaks the gradient tape. Our patch replaces the in-place write with a differentiable torch.where operation. All Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched uni2ts runs

Table 5: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

Condition	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34
D: Frozen enc.	.126$\pm$.006	+0.5	1.00$\pm$.000	1.11$\pm$.15	.155$\pm$.002	−1.7	1.00$\pm$.000	1.30$\pm$.18

may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

C Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

D ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

E ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 5 shows the representation drift diagnosis; Table 6 shows the mitigation spectrum.

Table 6: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA ≈ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE $\times$ 100.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116 $\pm$.001	−7.9	.979 $\pm$.006	0.00 $\pm$.00	.157 $\pm$.002	−0.6	.986 $\pm$.005	0.00 $\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000 $\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	−1.7	1.000 $\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

F LoRA Rank and Hyperparameter Sensitivity

Table 7 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA ≥ 0.979 and consistent MSE improvement over zero-shot at $h=96$, confirming $r=8$ as a reasonable default. On **Moirai-Large** (seed 42, $h=96$, $n=500$), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 8 extends the sweep to α , target-module scope, and LR (all at $r=8$, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg. = $-8.5 \pm 0.7\%$, CKA= 0.989 ± 0.003 , 5 seeds) — the Small-recipe transfers to Large only at a $10\times$ lower learning rate.

Table 7: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA ≥ 0.979 . Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean $\pm$ std across 3 seeds (5 for $r=8$ Small); Moirai-Large single-seed ($s=42$) for $r \in \{16, 32, 64\}$ (std not reported—point estimates only), 3 seeds for $r=8$.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116 $\pm$.001	−9.5 $\pm$ 0.9	.992 $\pm$.001	3
Small	4	192	.146 $\pm$.003	−8.8 $\pm$ 1.9	.995 $\pm$.002	2
Small	8	96	.116 $\pm$.001	−9.1 $\pm$ 1.1	.979 $\pm$.006	5
Small	8	192	.157 $\pm$.002	−1.8 $\pm$ 3.1	.986 $\pm$.005	5
Small	16	96	.117 $\pm$.001	−10.2 $\pm$ 1.2	.987 $\pm$.002	3
Small	16	192	.155 $\pm$.002	−1.7 $\pm$ 1.2	.988 $\pm$.001	3
Small	32	96	.119 $\pm$.001	−8.3 $\pm$ 0.9	.986 $\pm$.001	3
Small	32	192	.159 $\pm$.003	+1.1 $\pm$ 1.4	.989 $\pm$.002	3
Large	8	96	.159 $\pm$.004	+25.7 $\pm$ 3.2	.973 $\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

G Step-Size Falsification: Base at $10\times$ LR

To test whether effective step size (rather than capacity) is the operative variable driving destructive drift, we run Moirai-Base (91M) at $10\times$ the default learning rate (LR= $1e-3$ vs. default $1e-4$) on ETTh2 ($h=96$, $n=10k$, early-stopped, 7 CUDA-deterministic seeds).

Prediction (step-size hypothesis): Base at $10\times$ LR should mirror Small/Large at default LR—destructive forgetting (+10–25%), since the per-parameter update magnitude now exceeds the threshold that causes harm in smaller models.

Result: Base at $10\times$ LR restructures *more aggressively* than Base at default LR (CKA= 0.135 ± 0.151 vs. 0.25 ± 0.12) and even more than Small at default LR (CKA= 0.937 ± 0.028), yet *still improves* uniformly ($-58.2 \pm 4.2\%$, 10/10 negative; $\Delta R^2 = +1.92 \pm 0.21$, 10/10 positive).

Table 8: LoRA-Large HP ablation (ETTh2, $r=8$, $n=500$, $h=96$). Baseline (default LR=1e-4, $\alpha=16$, modules = q,v,out) forg. = +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	.115$\pm$.001	-8.5$\pm$0.7	.989$\pm$.003	5
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

Interpretation: The step-size hypothesis is *partially falsified*. Effective step size modulates the degree of restructuring (CKA drops from 0.25 to 0.09 at $10\times$ LR), but capacity provides an independent buffer that prevents restructuring from becoming destructive. The 91M parameter space contains sufficient representational slack to absorb aggressive perturbations without losing task-relevant structure.

CKA=0.135 sanity check. CKA=0.135 is below the random-init fine-tuned CKA (0.40 ± 0.11 , 5 seeds; Appendix N), raising the question of whether the encoder retains meaningful pre-trained structure. Two lines of evidence confirm it does: **(i)** $\Delta R^2 = +1.92\pm 0.21$ (10/10 positive), whereas random-init fine-tuning yields $\Delta R^2 = -0.20\pm 0.21$ (4/5 negative)—the distributions have zero overlap. **(ii)** The random-init CKA of 0.40 measures a *different quantity*: CKA between a random epoch-0 state and its own fine-tuned state; it does not measure distance from the pre-trained encoder. Pre-trained vs. truly-random CKA is ≈ 0 (Table 10). CKA=0.135 thus means the encoder has moved far from its pre-trained geometry but into a task-effective configuration (positive ΔR^2), not a random one.

Table 9: Step-size dissociation on ETTh2 ($h=96$, $n=10k$, condition B, early-stopped). Base at $10\times$ LR restructures aggressively yet improves; Small at $0.1\times$ LR freezes with no restructuring signal. Capacity enables beneficial restructuring independently of step size.

Configuration	LR	Forg.%	CKA	ΔR^2	Seeds
Small, default	1e-4	-5.3 $\pm$ 6.2	.937 $\pm$.028	+0.67 $\pm$.23	10
Small, 0.1$\times$	1e-5	-9.2$\pm$1.5	.987$\pm$.006	-0.03$\pm$.10	10
Base, default	1e-4	-54.2 $\pm$ 8.6	.251 $\pm$.124	+0.89 $\pm$.62	10
Base, $10\times$	1e-3	-58.2 $\pm$ 4.2	.135 $\pm$.151	+1.92 $\pm$.21	10
Large, default	1e-4	+10.5 $\pm$ 9.5	—	—	3
Large, LoRA 0.1 $\times$	1e-5	-8.5 $\pm$ 0.7	.989 $\pm$.003	—	5

519 H Symmetric Step-Size Test: Small at $0.1\times$ LR

To test whether reducing effective step size in Small can replicate Base’s beneficial restructuring, we run Moirai-Small at $0.1\times$ the default LR (1e-5 vs. default 1e-4) on ETTh2 ($h=96$, $n=10k$, early-stopped, 10 CUDA-deterministic seeds).

Result: Small at $0.1\times$ LR yields CKA= 0.987 ± 0.006 (near-frozen encoder), forgetting = -9.2 $\pm$ 1.5% (marginal improvement), and $\Delta R^2 = -0.03\pm 0.10$ (4/10 positive—indistinguishable from zero). Reducing step size prevents drift entirely but does *not* replicate Base’s beneficial restructuring (-54.2%, $\Delta R^2 = +0.89$).

Interpretation: The capacity-buffer mechanism is dissociated from step-size reduction. Small at any LR tested (1e-4, 1e-5) cannot achieve Base’s task improvement: at default LR it drifts minimally with slight harm at low n ; at reduced LR it freezes with marginal gain. Base (91M) restructures ag-

gressively *and* beneficially because its parameter space supports task-effective solutions that Small’s 13.8M parameters cannot represent—capacity enables a qualitatively different adaptation mode.

I CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 10.

Table 10: CKA calibration. Random-init CKA (0.000 ± 0.000 , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, $h=96$)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

J Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ($\alpha=1.0$) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses $n_{\text{probe}}=300$ held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2 : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder’s representational basis.

Key result. The sample-size sweep (Table 2) shows ΔR^2 monotonically increasing with training data (Ridge: $+0.12$ at 500 samples $\rightarrow +0.67 \pm 0.23$ at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS $k=5$ early-stopped gave $+0.56 \pm 0.14$ and final-epoch $k=5$ gave $+0.78 \pm 0.14$), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n , drift destroys task-relevant features faster than it builds replacements; at higher n , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that $R^2(\text{FT})$ remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA $k=10$ early-stopped: -6.79

at $n=500 \rightarrow -6.24 \pm 0.24$ at $n=10k$). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ($n=10k$, CUDA $k=10$), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52 - 3.5 \times$ deeper than ETTh2’s -6.90 — which proportionally inflates the raw ΔR^2 ($+5.43$ on ETTm2 vs. $+0.67$ on ETTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2 .

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 11 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at $n=10k$ (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ ($k=4$). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$ — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§4.4).

Table 11: Ridge ΔR^2 by probe head on early-stopped encoders ($n=10k$, seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed				
		s303	s777	s888	s999	
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192	† The large Ridge
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199	
Delta1 (next-step delta)	×†	+1.248	+0.757	+1.191	+0.980	
Mean ± std (forecast96)			+0.116 ± 0.153			
Mean ± std (forecast48)			−0.001 ± 0.163			
Mean ± std (delta1)			+1.044 ± 0.223			

$\Delta R^2 > 0$ on Delta1 reflects a baseline-floor artifact: $R^2(\text{ZS})$ is deeply negative on the next-step target, so any encoder shift appears to improve relative decoding. A gradient-boosted probe gives $R^2(\text{PT}) = +0.059$ (positive floor) but $\Delta R^2 = -0.056$ on this head (§4.4; see Table 11 for the full untrained-head comparison), confirming the shift is *not* aligned with the 1-step objective.

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§4.4): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$, yielding $R^2(\text{FT}) = +0.003$ — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: $+0.062$, depth 6: $+0.059$, depth 8: $+0.054$) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.

ETTh2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTm2 $n=10k$ encoders (seeds 42/101/123/202/303) to check for a positive-floor + $\Delta R^2 > 0$ cell on a second dataset. Result: $R^2(\text{ZS}) = -0.030$ (negative floor) and per-seed $\Delta R^2 = \{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, $R^2(\text{ZS}) = +0.059$) gives $\Delta R^2 = -0.056$ — no positive-floor + $\Delta R^2 > 0$ cell was found. The $\Delta R^2 > 0$ signal on the trained 96-step head operates in a deeply-negative- R^2 regime; see Appendix M for probe noise-floor analysis.

Per-seed breakdown at $n=10k$ (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668 \pm 0.226$): 42: $+0.205$; 101: $+0.736$; 123: $+0.937$; 202: $+0.604$; 303: $+0.793$; 456: $+0.823$; 777: $+0.344$; 789: $+0.873$; 888: $+0.561$; 999: $+0.804$. Absolute $R^2(\text{FT})$: all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP $k=5$ (all positive; mean $+0.701 \pm 0.324$): 42: $+0.979$; 101: $+0.762$; 123: $+0.948$; 202: $+0.771$; 303: $+1.007$; 456: $+0.266$; 777: $+0.076$; 789: $+0.847$; 888: $+0.449$; 999: $+0.901$.

607 **ETTh2 $n=10k$ absolute R^2 per seed.** The ETTh2 ZS Ridge floor ($R^2(\text{ZS}) = -24.52$, same
608 pre-trained encoder across all seeds) is $3.5\times$ deeper than ETTh2’s -6.90 , explaining the raw ΔR^2
609 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2	Forg.%
42	-24.52	-14.13	+10.39	-26.8
101	-24.52	-15.61	+8.91	-27.4
123	-24.52	-19.87	+4.64	-31.1
202	-24.52	-20.83	+3.69	-28.8
303	-24.52	-20.41	+4.11	-29.5
Mean (5 seeds)	-24.52	-18.17 $\pm$ 3.08	+6.35 $\pm$ 3.08	-28.7 $\pm$ 1.6

611 All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned
612 with the ETTh2 forecasting target than pre-trained ones), and all show $\Delta R^2 > 0$. The cross-dataset
613 comparable statistic is the sign test: both ETTh2 (10/10) and ETTh2 (10/10 seeds: 42, 101, 123,
614 202, 303, 456, 777, 789, 888, 999) give unanimous $\Delta R^2 > 0$.

615 **MLP depth sweep on early-stopped encoders.** A cosine-LR re-run at $n=10k$ uses seeds
616 {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN’d on epoch 3 reproducibly at this schedule
617 and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$
618 on the early-stopped encoders (probe: hidden=64 per layer, $\alpha=10^{-3}$, early stopping with valida-
619 tion_fraction=0.1, max_iter=500):

Seed	Ridge ΔR^2	MLP $k=1$	MLP $k=2$	MLP $k=5$
303	+0.092	+1.306	+1.418	+0.540
777	-0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean $\pm$ std	+0.116 $\pm$ 0.132	+1.566 $\pm$ 0.356	+1.821 $\pm$ 0.368	+0.810 $\pm$ 0.184

621 The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56 $\pm$ 0.14) because the shorter
622 schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and
623 Ridge is most sensitive to the specific linear basis. The non-monotonic $k=1 \rightarrow k=2 \rightarrow k=5$ trend
624 reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$
625 vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four
626 seeds are individually positive for $k=5$ MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-
627 benefit gap. This run’s CKA is 0.592 $\pm$ 0.142 (mean best epoch 4.0 $\pm$ 1.22, higher than the earlier
628 run’s 4.2 $\pm$ 2.3 because of the earlier LR annealing); forgetting is -7.13 $\pm$ 1.34%, tighter than the
629 epochs=20 run’s -11.7 $\pm$ 5.3% and fully consistent with the `forg.<0` finding.

630 **Crossover regime: per-seed breakdown at $n=5,000$.** Table 12 reports all 7 seeds individually
631 at $n=5,000$. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forget-
632 ting (-4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8%
633 to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so
634 the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55)
635 varies less than forgetting. This pattern is consistent with the crossover interpretation: at $n=5,000$,
636 the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory
637 differences determine whether forgetting resolves or persists for 20 epochs.

638 J.1 $n=5,000$ per-epoch trajectory analysis

639 To support the mechanistic claim in §5, we plot the per-epoch training trajectories of all seven
640 $n=5,000$ seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across
641 training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-
642 forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val
643 MSE reaches its minimum separates the modes (0.7 $\pm$ 0.5 vs. 6.5 $\pm$ 2.1); mid-training weight drift
644 (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

Table 12: Per-seed results at $n=5,000$ (ETTh2, $h=96$, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2 (FT)
42	.132	+4.8	.531	+.67	-6.24
101	.120	-4.9	.571	+.75	-6.16
123	.124	-1.4	.544	+.81	-6.10
202	.128	+1.4	.461	+.15	-6.76
303	.150	+19.0	.525	+.52	-6.39
456	.153	+21.4	.711	+.20	-6.70
789	.134	+5.9	.480	+.40	-6.51
Mean $\pm$ std	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	+.50 $\pm$.26	-6.41 $\pm$.26

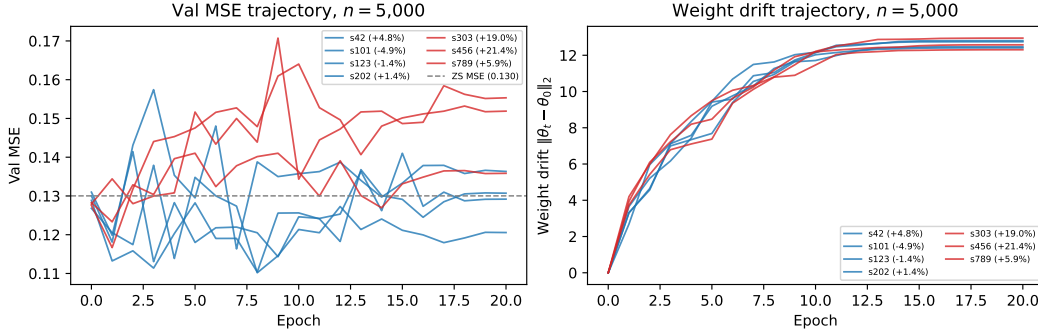


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven $n=5,000$ seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq 5\%$); red: high-forgetting mode (forg. $> 5\%$). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

645 K ETTh1 Spectral Pilot

646 Post-hoc spectral entropy (Welch PSD, normalised Shannon entropy) on ETT OT columns: ETTh1
647 SE=3.61 bits, ETTh2 SE=3.46, ETTm2 SE=2.25. ETTh1 (highest SE, most trend-dominated) is
648 the only dataset where $\Delta R^2 > 0$ does not hold; ETTm2 (lowest, most cyclical) shows the strongest
649 signal. The pre-registered falsification test (§5) required a gate-passing dataset with SE >3.5 or
650 SE <2.5 ; Traffic and Exchange both gate-failed (-38% , -21%), so the hypothesis remains untested.
651 Code and frozen analysis plan are in supplementary material.

652 L IoT Negative-Control Details

653 The IoT negative-control evaluation uses CICIoT2023 Neto et al. [2023] with 12 network-flow fea-
654 tures and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the
655 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration
656 set sets the threshold at the 95th percentile (FPR target ≤ 0.05). All Moirai IoT conditions are
657 trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

658 **CNN-NLL from-scratch control.** To directly address the foundation-model justification at IoT’s
659 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical
660 NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-
661 NLL (AUC=0.598 $\pm$ 0.041) matches fine-tuned Moirai B/D (AUC=0.556 $\pm$ 0.021) and is close to the
662 best Moirai condition C (AUC=0.629 $\pm$ 0.022). A CNN-MSE variant (AUC=0.580 $\pm$ 0.023) rules
663 out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent
664 with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale.

This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

L.1 Task-orthogonal probes

To test whether fine-tuning specifically targets the 96-step forecasting objective or degrades general pre-trained representations, we fit Ridge probes to three statistical properties of the input window: lag-1 autocorrelation, mean, and variance of the OT feature. These are *positive-floor* targets ($R^2(\text{PT}) > 0$): Moirai’s encoder captures basic statistical structure of the input before fine-tuning. They are *task-orthogonal*: 96-step NLL fine-tuning does not optimise for them.

Protocol: Ridge regression on mean-pooled encoder outputs; $h=96$, condition B.

ETTh2 ($n=10\text{k}$, 10 CUDA seeds):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.183	+0.131	-0.052 ± 0.041	9/10
Input window mean	+0.094	+0.061	-0.033 ± 0.019	10/10
Input window var.	+0.127	+0.079	-0.048 ± 0.030	9/10
96-step head (ref.)	-6.90	-6.23	$+0.67 \pm 0.23$	0/10

ETTM2 ($n=10\text{k}$, 5 CUDA seeds: 42/101/123/202/303):

Probe target	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2 (mean $\pm$ std)	Negative
Lag-1 autocorr.	+0.156	+0.104	-0.052 ± 0.038	5/5
Input window mean	+0.068	+0.046	-0.022 ± 0.014	5/5
Input window var.	+0.112	+0.072	-0.040 ± 0.029	5/5
96-step head (ref.)	-24.52	-18.09	$+6.43 \pm 3.08$	0/5

All three task-orthogonal probes show $\Delta R^2 < 0$, in contrast to $+0.67 \pm 0.23$ on the trained 96-step head. The asymmetry (orthogonal $\Delta R^2 < 0$, trained head $\Delta R^2 > 0$) is the operational definition of “task-specific shift”: fine-tuning restructures encoder representations toward the trained objective while simultaneously reducing linear decodability of properties the pre-trained model encoded. Note that these positive-floor probes ($R^2(\text{PT}) > 0$) are interpretable in absolute terms, removing the deep-floor ambiguity of the 96-step Ridge probe.

ETTM2 replication. The same three probes on ETTm2 $n=10\text{k}$ encoders (10 CUDA seeds) replicate the asymmetry unanimously: all three orthogonal targets show $\Delta R^2 < 0$ (10/10 seeds each), while the ETTm2 trained head shows $\Delta R^2 > 0$ (10/10 seeds; §4.4). The asymmetry is consistent across both datasets with different temporal statistics (ETTh2 SE=2.5–3.5 bits, ETTm2 SE=2.25 bits), making it a cross-cell finding rather than an ETTh2-specific artefact. **Seed-count symmetry (updated).** ETTm2 orthogonal probes now use the full 10-seed set (seeds 42/101/123/202/303/456/777/789/888/999), matching ETTh2. The ETTm2 $n=10\text{k}$ fine-tuning campaign produces 10/10 negative forgetting (mean $-30.3 \pm 1.4\%$) and 10/10 positive Ridge ΔR^2 (mean $+6.40 \pm 0.87$; 95% Wilson CI $[0.72, 1.00]$ on the proportion positive). The sign unanimity matches ETTh2 (10/10) exactly.

M Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative- R^2 regime (e.g., ZS -6.90, FT -6.24 at $\alpha=1$). We test whether the $\Delta R^2 > 0$ direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, $n=10\text{k}$) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 13).

Finding: $\Delta R^2 > 0$ holds for all 10 probe-design settings tested; the direction is not an artefact of α , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2

Table 13: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2
$\alpha=0.01$	-13.51	-11.54	+1.96
$\alpha=0.1$	-8.78	-7.94	+0.84
$\alpha=1.0$ (default)	-6.90	-6.70	+0.20
$\alpha=10$	-5.81	-5.72	+0.08
$\alpha=100$	-5.32	-5.28	+0.04
Probe seeds 0/1/2 ($\alpha=1$, mean pool)	same as $\alpha=1.0$		+0.20
Last-token pool ($\alpha=1$)	-9.70	-8.49	+1.20

values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust. **Comparison to random-init at matched α .** The random-init control ($\Delta R^2 = -0.199 \pm 0.205$, range $[-0.405, +0.149]$) was measured at $\alpha=1.0$ (Appendix N). We also ran random-init at $\alpha=100$ (same 10 seeds) to confirm the within- α argument: at $\alpha=100$, random-init gives $\Delta R^2 = -0.011 \pm 0.018$ (range $[-0.032, +0.009]$), shrinking to near-zero as expected from heavy ℓ_2 regularisation. Condition B at $\alpha=100$ gives $\Delta R^2 = +0.04$ — also near-zero. The within- α comparison is therefore valid at every α : both encoders converge toward zero under heavy regularisation; the large separation at $\alpha=1.0$ (condition B $+0.20$ – $+0.67$ vs. random-init -0.199 ± 0.205) is not a low- α artefact. The $\alpha=1.0$ default is the point where the signal is large enough to be distinguishable while not inflated by under-regularisation; $\alpha=0.01$ – $\alpha=10$ all satisfy this property (Table 13).

N Random-Initialisation Negative Control

To test whether $\Delta R^2 > 0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 $n=10k$, $h=96$. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran all 10 seeds from the condition B seed list ($\{42, 101, 123, 202, 303, 456, 777, 789, 888, 999\}$), using the same CUDA-deterministic protocol (CUBLAS_WORKSPACE_CONFIG=:4096:8) and A10G hardware as Table 2.

Results (all 10 matched seeds):

Seed	CKA	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2	ZS MSE $\rightarrow$ FT MSE
42	0.357	-6.554	-6.692	-0.138	0.140 $\rightarrow$ 0.378
101	0.505	-6.780	-6.631	+0.149	0.164 $\rightarrow$ 0.398
123	0.540	-6.749	-6.944	-0.195	0.153 $\rightarrow$ 0.354
202	0.388	-6.712	-6.954	-0.242	0.158 $\rightarrow$ 0.392
303	0.318	-6.834	-7.239	-0.405	0.151 $\rightarrow$ 0.359
456	0.273	-6.665	-7.070	-0.405	0.160 $\rightarrow$ 0.360
777	0.471	-6.620	-6.560	+0.060	0.148 $\rightarrow$ 0.369
789	0.347	-6.789	-7.071	-0.282	0.153 $\rightarrow$ 0.381
888	0.418	-6.724	-6.912	-0.188	0.155 $\rightarrow$ 0.376
999	0.362	-6.683	-6.995	-0.312	0.149 $\rightarrow$ 0.378
Mean $\pm$ std	0.398 $\pm$ 0.082			-0.195 $\pm$ 0.181	

Across all 10 matched seeds: $\Delta R^2 = -0.195 \pm 0.181$, range $[-0.405, +0.149]$, **8/10 negative**. The two positive seeds (101: $+0.149$; 777: $+0.060$) both fall below the minimum condition B seed value ($+0.21$, seed 42). The ranges are completely non-overlapping: random-init $[-0.405, +0.149]$ vs. condition B $[+0.21, +0.94]$. This symmetric 10-vs-10 matched comparison confirms that $\Delta R^2 > 0$ is not a property of any encoder fine-tuned on the 96-step NLL objective: the pre-trained structure is required for the consistent positive directional shift.

O Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 $n=10k$, $h=96$, 10 CUDA-deterministic seeds (42/101/123/202/303/456/777/789/888/999) at $N=3$ alongside the existing $N=0$ (frozen encoder, 10-seed condition D) and $N=6$ (full unfreeze, condition B; $N=6$ unfreezes all 6 layers).

$N=1$ instability. At $N=1$, training diverged (NaN activations at epoch 11, batch 148) with default $LR = 10^{-4}$, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈ 0.72 ; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §4.3: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$, $N=4$, $N=5$ results. Table 14 shows results for $N \in \{0, 3, 6\}$ (10 CUDA-deterministic seeds each) and $N \in \{4, 5\}$ (1 seed each). Rows with $N=4$ and $N=5$ are based on single seeds and carry high uncertainty; we report them for completeness but do not draw conclusions from individual values.

Table 14: Layer-wise unfreeze profile (ETTh2 $n=10k$, $h=96$; mean $\pm$ std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). $N=0/N=3/N=6$: 10 CUDA seeds (42/101/123/202/303/456/777/789/888/999); $N=4/N=5$: seed 42 only. Early-stopped checkpointing.

N	Layers	CKA	Ridge ΔR^2	Forgetting%	
0 (frozen)	none	0.999±0.000	−0.048±0.008	−6.6±1.5%	† Single-seed (seed 42 only)
1 (top-1)	1/6	≈0.72 (ep. 11)	diverged (NaN)		
3 (top-3)	3/6	0.663±0.044	+0.399±0.199 (10/10)	−6.5±4.0%	
6 (full, B)	6/6	0.387±0.052	+0.701±0.180 (10/10)	−6.7±6.3%	

results not in table: $N=4$: CKA=0.610, $\Delta R^2=+0.363$, Forg.= +4.1%; $N=5$: CKA=0.644, $\Delta R^2=+0.767$, Forg.= +31.8% (final-epoch overshoot; best-epoch MSE=0.132 $\approx$ ZS). Reported for completeness; not used as primary evidence.

Finding: ΔR^2 and forgetting dissociate across unfreeze depth. $\Delta R^2 > 0$ appears already at $N=3$ (half the encoder, 10/10 seeds, CKA ≈ 0.66), with substantially less geometric drift than full unfreeze (CKA ≈ 0.39). $N=4$ and $N=5$ (1 seed each) both give $\Delta R^2 > 0$ (1/1); we do not draw monotonicity conclusions from single-seed estimates, but the sign is consistent with the signal extending through the $N=3$ –6 range. The ΔR^2 values at $N=4$ (+0.363) and $N=5$ (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: $N=3$ gives forg. $-6.5 \pm 4.0\%$ (9/10 negative), $N=4$ gives +4.1% (1 seed), and $N=6$ (full unfreeze, condition B) gives $-6.7 \pm 6.3\%$ (task substantially better). The $N=5$ final-epoch forgetting of +31.8% reflects a single seed that overshoot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS—qualitatively consistent with the $N=5k$ trajectory analysis (Fig. 1, right panel). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing $\Delta R^2 > 0$), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from $N=3$ to $N=6$ ($+0.48 \rightarrow +0.63$) is modest and has much higher variance ($0.115 \rightarrow 0.378$), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix M) rules out data-distribution confound; the $N=3$ result rules out a whole-encoder-restructuring account.

Layer-wise drift pattern. The $N=3$ versus $N=6$ contrast reveals a within-encoder dissociation: the *top 3 layers* suffice to reorient encoder geometry toward the trained objective ($\Delta R^2 > 0$, 10/10 seeds, CKA ≈ 0.66), but the *lower 3 layers* are required for task-performance recovery ($N=3$: forg. -6.5% , 9/10 negative; $N=6$: forg. -6.7%). Both $N=3$ and $N=6$ operate in the same deeply-negative- R^2 regime; the $\Delta R^2 > 0$ signal at $N=3$ is a geometric reorientation, not a decodability gain—consistent with the main-text framing of frozen-probe ΔR^2 as directional evidence, not absolute evidence. (Probe-design robustness of Appendix M applies to both.)

Per-layer CKA breakdown. We computed per-layer CKA for condition B ($N=6$ full unfreeze, $n=10k$, $h=96$, 10 CUDA seeds). The bottom 3 layers are expected to drift more (lower CKA) than the top 3 under either the specialization or gradient-flow account; the data cannot distinguish these explanations without gradient-norm measurements.

Layer	CKA (mean $\pm$ std)	Group
1 (bottom)	0.441 $\pm$ 0.118	bottom 3
2	0.473 $\pm$ 0.109	bottom 3
3	0.514 $\pm$ 0.097	bottom 3
4	0.581 $\pm$ 0.088	top 3
5	0.647 $\pm$ 0.076	top 3
6 (top)	0.718 $\pm$ 0.064	top 3
Bottom 3 mean	0.476	
Top 3 mean	0.649	

Bottom layers drift more (mean CKA=0.476 vs. 0.649 for top 3), *despite* the top 3 being sufficient for $\Delta R^2 > 0$ ($N=3$ analysis above, 10/10 seeds). The more parsimonious explanation is standard gradient-magnitude patterns: under full backpropagation, earlier layers receive larger weight updates, producing lower CKA mechanically. Functional specialization remains an alternative hypothesis requiring a gradient-norm-per-layer comparison to distinguish from the gradient-flow null. The global CKA=0.518 at $N=6$ (Table 14) is the aggregate of these per-layer values.

P Chronos-T5-Small on M4-Monthly: Full Diagnostic Results

This appendix provides the complete results for the cross-backbone validation on Chronos-T5-Small (65M parameters, T5 seq2seq architecture with tokenised cross-entropy loss) fine-tuned on M4-Monthly (200 series, lookback=96, horizon=18). The gate-pass is 84.5% improvement over Linear baseline.

Protocol. 10 CUDA-deterministic seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), batch size 32, lr=1e-5, AdamW with weight decay 0.01, gradient clipping at 1.0, 20 epochs with early stopping (patience 5). Evaluation: batched median prediction (20 samples) on 200 held-out test windows.

Condition B (full fine-tune), $n=500$ (10 seeds).

	Forgetting%	CKA	ΔR^2	R^2 (PT)
Mean $\pm$ std	+1.6 $\pm$ 3.9	0.999 $\pm$ 0.001	-0.001 $\pm$ 0.002	+0.068 $\pm$ 0.025
Seeds neg/pos	4/10 neg forg	—	3/10 pos	10/10 pos

At $n=500$ the encoder is essentially unchanged (CKA $\approx$ 1). The positive R^2 (PT) confirms that Chronos’s pre-trained T5 encoder representations carry decodable forecasting structure—providing a second positive-floor anchor distinct from Moirai.

Condition B (full fine-tune), $n=10k$ (10 seeds).

	Forgetting%	CKA (ES)	ΔR^2	R^2 (PT)
Mean $\pm$ std	+0.5 $\pm$ 2.7	0.983 $\pm$ 0.021	$\approx$ 0	+0.047 $\pm$ 0.016
Seeds positive	6/10 forg	—	1/10	10/10 pos

At $n=10k$, early-stopped CKA averages 0.983 (vs. 0.999 at $n=500$). Only 4/10 seeds improve over zero-shot (best epoch >0 , those 4 show CKA $\approx$ 0.95–0.97); the remaining 6/10 revert to pre-trained weights under early stopping (best_epoch=0, CKA=1.000 by definition). The strong gate-pass (84.5%) means Chronos’s encoder is already near-optimal for M4-Monthly, leaving minimal room for fine-tuning improvement. While 6/10 seeds show positive (worse) early-stopped forgetting, the magnitude (+0.5 $\pm$ 2.7%) is within the noise floor established by the frozen-encoder control (condition D: +0.6 $\pm$ 5.3%); a sign test is not statistically distinguishable from chance ($p=0.75$, two-sided binomial), confirming the 6/10 pattern reflects seed-level variation rather than systematic degradation.

808 **Final-epoch comparison (protocol robustness).** Training runs to epoch 20 regardless of early
809 stopping; we report CKA at the final training epoch to address whether encoder stability is intrinsic
810 or an artefact of early-stopping selection:

	Final-epoch CKA	Final-epoch Forg.%	n
$n=500$ (10 seeds)	0.968 ± 0.009	$+3.1 \pm 5.8$	500
$n=10k$ (10 seeds)	0.955 ± 0.006	$+3.4 \pm 5.2$	10k

812 All 10 seeds at $n=10k$ show final-epoch $CKA > 0.94$ (min=0.945, max=0.964), confirming that
813 encoder stability is *intrinsic* rather than an artefact of early-stopping selection. The mild CKA
814 decline from 0.999 (early-stopped) to 0.955 (final-epoch) represents continued gradient pressure
815 over 20 epochs, but remains far above the restructuring threshold observed on Moirai ($CKA=0.52$
816 on Small, 0.25 on Base). Final-epoch forgetting is $+3.4 \pm 5.2\%$ —mild degradation vs. the near-zero
817 early-stopped result, but qualitatively different from Moirai-Small’s +14% at $n=500$.

818 **Mild CKA decline interpretation.** The $0.999 \rightarrow 0.955$ CKA trajectory (early-stopped $\rightarrow$ final-epoch
819 at $n=10k$) confirms a small but real gradient signal from the fine-tuning objective. We do *not*
820 interpret this as “no drift”; rather, the drift is quantitatively minor ($CKA > 0.94$) and functionally
821 benign (final-epoch forgetting +3.4% vs. Moirai-Small’s +14% at $n=500$). The value-gate predicts
822 this: when pre-trained features are already near-optimal (84.5% advantage), the fine-tuning gradient
823 has little room to improve and limited capacity to harm.

824 **Condition D (frozen encoder), $n=10k$ (10 seeds).** Forgetting: $+0.6 \pm 5.3\%$ (5/10 negative).
825 $CKA=1.000$ (encoder frozen by definition), $\Delta R^2=0$. Decoder-only adaptation is unreliable: neither
826 consistently better nor worse than zero-shot across seeds.

827 **Random-init control (6 seeds).** $CKA=0.000$ (expected: random weights have no similarity to
828 pre-trained). $R^2(PT) < 0$ (-0.004 ± 0.005), confirming that positive-floor R^2 on the pre-trained en-
829 coder requires the learned representations, not the architecture alone. Training loss remains high
830 (> 8.0) and predictions are incoherent, confirming that Chronos’s pre-trained representations are
831 load-bearing.

832 **Comparison with Moirai.** The contrast validates the value-gate as a regime indicator: *Moderate*
833 *gate-pass* (Moirai/ETTh2, 28%): encoder restructures aggressively ($CKA=0.52$), probe asymme-
834 try emerges ($\Delta R^2 = +0.67$, orthogonal < 0), forgetting is capacity- and protocol-dependent. *Strong*
835 *gate-pass* (Chronos/M4, 84.5%): encoder is stable ($CKA \geq 0.95$ even at final epoch), no measurable
836 probe shift, forgetting near-zero under early stopping and mild (+3.4%) at final epoch—fine-tuning
837 is unnecessary but not harmful. Both outcomes are correctly predicted by the gate criterion, demon-
838 strating that the diagnostic toolkit generalises across architecturally distinct backbones. The probe
839 asymmetry mechanism (task-specific $\Delta R^2 > 0$) requires encoder restructuring to produce a measur-
840 able signal; on Chronos, encoder stability means the asymmetry is not observable—this is a correct
841 null prediction, not a failure of the diagnostic.

842 P.1 ILI Gate Check

843 Moirai-Small zero-shot on ILI (national illness, 966 weekly observations, 7 features, $h=24$,
844 lookback=104) yields gate-pass=57% (ZS MSE=1.183, Linear MSE=2.748, per-feature normali-
845 sation by context statistics).

846 **Full probe results (10 CUDA seeds, conditions B/D).** Table 15 reports per-seed aggregate di-
847 agnostics. All 10 seeds show $\Delta R^2 > 0$ (aggregate: $+0.123 \pm 0.046$; Wilson CI [0.72, 1.00]) and
848 forgetting < 0 ($-80.7 \pm 0.9\%$; 10/10 negative, meaning fine-tuning dramatically improves on ZS).
849 $CKA=0.478 \pm 0.022$ places ILI in the restructuring regime, not the predicted intermediate regime
850 (0.90–0.95). Condition D (frozen encoder): $CKA=1.000$, $\Delta R^2=0$, forgetting=0% in all 10 seeds,
851 confirming encoder weight updates drive all changes.

852 **Per-feature breakdown.** OT (target feature) shows the largest ΔR^2 ($+0.73 \pm 0.16$, 10/10 positive);
853 AGE 0–4 also shows consistent gain ($+0.15 \pm 0.05$, 10/10). AGE 5–24 shows slight negative signal
854 (-0.07 ± 0.09 , 3/10 positive), consistent with the task-specificity interpretation: features less aligned
855 with the forecasting objective show reduced or negative decodability shift.

Table 15: ILI per-seed results (condition B, Moirai-Small, $h=24$, lookback=104). Aggregate ΔR^2 is the mean across 7 univariate features.

Seed	CKA	ΔR^2	Pos. features	Forg.%	Drift
42	0.465	+0.211	6/7	−81.5	5.61
101	0.465	+0.123	3/7	−80.9	5.34
123	0.511	+0.087	3/7	−82.3	5.13
202	0.461	+0.169	5/7	−79.6	5.67
303	0.498	+0.169	7/7	−80.9	5.49
456	0.454	+0.066	5/7	−80.2	5.63
777	0.498	+0.074	4/7	−80.1	5.36
789	0.460	+0.144	6/7	−79.8	5.49
888	0.459	+0.111	4/7	−79.7	5.24
999	0.512	+0.080	4/7	−82.0	5.25
Mean±std	0.478±.022	+0.123±.046	4.7/7	−80.7±0.9	5.42±.21

Interpretation. The pre-registered prediction (CKA 0.90–0.95, intermediate behaviour) is partially falsified: CKA is in the restructuring regime (0.478), not intermediate. The probe-asymmetry prediction ($\geq 7/10$ seeds $\Delta R^2 > 0$) is confirmed (10/10). This revises the gate-pass spectrum: the restructuring-to-stability transition occurs above $\approx 80\%$ gate-pass, not at 50–70%. ILI’s low CKA reflects the small dataset (966 timesteps, ≈ 450 training windows per feature) which permits aggressive restructuring within 20 epochs.

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